

Project - Low Level Design On ProjectTitle

Sports Content Generator

Course Name: Generative AI

Institution Name: Medicaps University – Datagami Skill Based Course

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1 Introduction

The Low Level Design (LLD) describes the detailed internal implementation of the Sports Content Generator system. It explains how each module functions, how data flows between components, and how the application processes user requests to generate structured sports content using a Large Language Model (LLM) API. This design ensures modularity, scalability, maintainability, and secure API communication.

The system is divided into the following internal modules:

1.1 User Interface Module (Streamlit Layer)

Technology Used: Python, Streamlit

Responsibilities:

- Capture user input (sports topic and content type)
- Provide interactive UI components (text box, dropdown, buttons)
- Display generated sports content
- Display error messages

Inputs:

- Topic (string)
- Content Type (string)

Outputs:

- Generated sports article
- Error messages (if any)

1.2 Input Validation Module

Ensures only valid data is sent to the content generation module.

Validations Performed:

- Empty input check
- Minimum and maximum length check
- Valid content type selection
- Basic input sanitization

Outcome:

- Returns validation status (True/False)
- Error message (if validation fails)

1.3 Content Generation Module

Purpose:

Handles prompt creation, content generation, and formatting.

Internal Responsibilities:

- Construct structured prompt
- Call API service module
- Process raw API response
- Format content into readable structure

Core Methods:

- generate_prompt(topic, type)
- generate_content(topic, type)
- format_output(response)

Purpose:

Handles secure communication with the Large Language Model API.

Responsibilities:

- Manage API key via environment variables
- Send HTTP request
- Handle API response
- Manage API errors and timeouts

Security Measures:

- API key not hardcoded
- HTTPS communication
- Exception handling for failures

2.Class Design

Class: StreamlitUI

Methods:

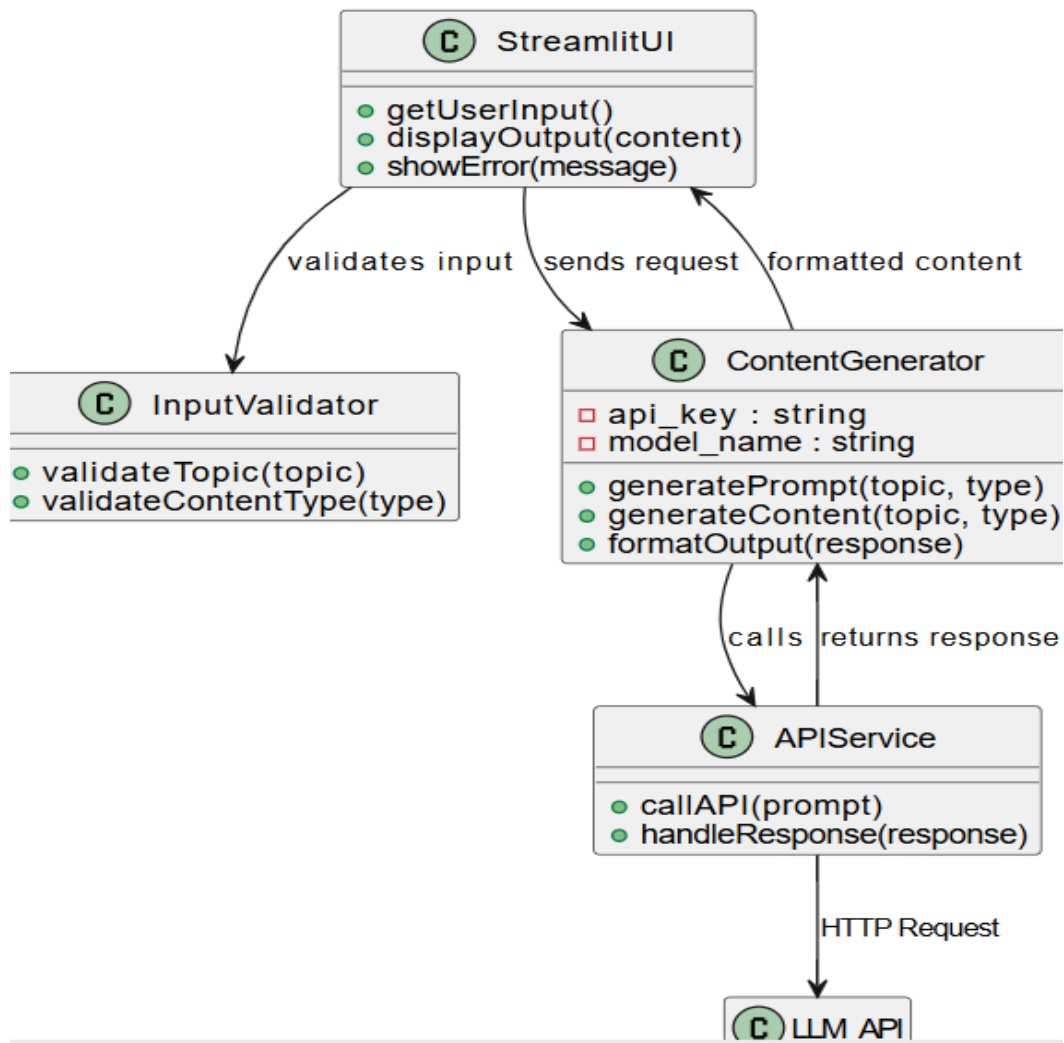
- `get_user_input()`
- `display_output(content)`
- `show_error(message)`

Class: InputValidator

Methods:

- `validate_topic(topic)`
- `validate_content_type(type)`

Sports Content Generator - Low Level Design (Class Diagram)



Class: ContentGenerator

Attributes:

- `api_key : string`
- `model_name : string`

Methods:

- `generate_prompt(topic, type)`
- `generate_content(topic, type)`

- `format_output(response)`

Class: APIService

Methods:

- `call_api(prompt)`
- `handle_response(response)`

3.Detailed Process Flow

1. User enters topic in Streamlit interface
2. User selects content type
3. User clicks "Generate" button
4. InputValidator validates user input
5. If validation fails → Error displayed
6. If validation succeeds → ContentGenerator constructs prompt
7. APIService sends request to LLM API
8. API returns generated sports content
9. ContentGenerator formats response
10. Streamlit displays formatted article